

Jalostus - Generic Model of Value Addition

A General Method for Analyzing Organizational Structure, Operations and Value Creation

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Model scope:	Industry-independent
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Abstract

THE GENERIC MODEL OF VALUE ADDITION IS AN INDUSTRY-INDEPENDENT FRAMEWORK FOR DESCRIBING AND ANALYZING ORGANIZATIONAL STRUCTURE, ECONOMIC ACTIVITY, AND VALUE CREATION. THE MODEL CONCEPTUALIZES THE ORGANIZATION AS A LARGE ECONOMIC SUMMARIZER, WITH EACH FUNCTION OPERATING AS AN ECONOMIC SUMMARIZER WHOSE RESULT IS DETERMINED BY INCOME, PURCHASE COSTS, AND REFINING. THUS, ORGANIZATIONAL PROFIT/LOSS IS REPRESENTED AS THE AGGREGATE OF FUNCTION-LEVEL ECONOMIC RESULTS.

THE MODEL INTEGRATES ORGANIZATIONAL FUNCTIONS WITH PRODUCT BUILD SCOPE AREAS TO SUPPORT ANALYSIS OF RESOURCE ALLOCATION, VALUE CREATION, AND PROFITABILITY ACROSS DIFFERENT LEVELS OF ORGANIZATIONAL ACTIVITY. IT DOES NOT REQUIRE INDIVIDUAL FUNCTIONS TO BE INDEPENDENTLY PROFITABLE; THEIR ECONOMIC CONTRIBUTION MAY ARISE THROUGH THEIR ENABLING EFFECT ON OTHER FUNCTIONS AND ON THE ORGANIZATION AS A WHOLE.

ORIGINALLY DEVELOPED FROM OBSERVATIONS OF IT SERVICE BUSINESS FROM 1999 ONWARD AND FIRST DOCUMENTED IN 2001, THE MODEL WAS FORMALIZED IN 2010 AS A GENERAL ORGANIZATIONAL ANALYSIS METHOD. THE ORIGINAL 2001–2003 PLANNING SCENARIOS FURTHER DEMONSTRATE ITS USE AS AN ITERATIVE MANAGEMENT TOOL: ORGANIZATIONAL ECONOMICS CAN BE DESCRIBED, SUMMARIZED, MODIFIED, AND RE-EVALUATED TO IDENTIFY POTENTIAL IMPROVEMENTS. THESE SCENARIOS ARE ILLUSTRATIVE AND SPECULATIVE RATHER THAN HISTORICAL ACCOUNTING DATA.

THE MODEL PROVIDES A BASIS FOR SYSTEMATIC ANALYSIS OF THE RELATIONSHIP BETWEEN ORGANIZATIONAL STRUCTURE, RESOURCE CONSUMPTION, REFINING, VALUE ADDITION, AND OVERALL PROFITABILITY, AND OFFERS A CONCEPTUAL FOUNDATION FOR FUTURE COMPUTATIONAL APPLICATIONS USING ACCOUNTING AND ORGANIZATIONAL DATA.

1 Introduction

1.1 Background

Jalostus by T. Lind, or the **Generic Model of Value Addition**, was developed from observations made in information-technology service business from 1999 onward. The model was first presented in written form to Fujitsu-Siemens Professional Services Finland business management on 15 August 2001 under the title *Hosting Business*.

The original purpose was to describe an organization's functions, their dependencies and their economic relationships.

The model subsequently developed into a more general method for describing and analyzing organizational structure, activities, costs, value creation and profitability.

1.2 General principle

Jalostus by T. Lind treats an organization as a collection of **functions**.

A function is an organizational area in which activities are performed. The model does not require a particular organizational structure: an organization may contain any selection and number of the model's functions.

The model is intended to be independent of what the organization produces. It can therefore be applied to any organization whose activities can be analyzed in terms of resources, activities, value and economic flows, including non-profit and non-profitable organizations.

1.3 Model perspective

The central perspective is not the physical nature of an output but the **value-adding activity performed by the organization**.

The organization can consequently be viewed as one large summarizer:

- economic input enters from above,
- individual functions perform their activities,
- funds are consumed by Purchase Costs and Refining,
- function-level results are summarized,
- and the organization's resulting Profit/Loss appears at the bottom.

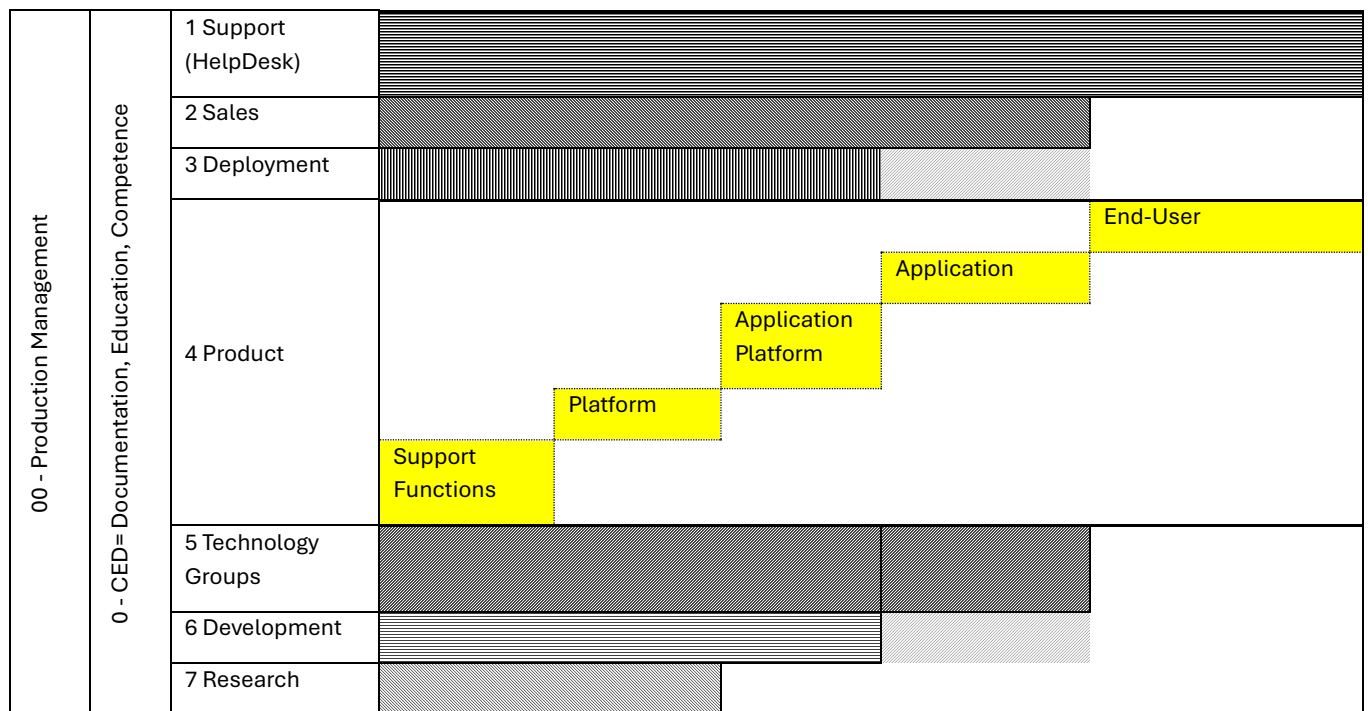
A negative result can correspond to a need for funding from the bottom rather than profit flowing out.

2 Organization and Functions

2.1 Canonical function structure

The functions are maintained in the following order:

No.	Function
00	Management
0	DEC — Documentation, Education, Competence
1	Customer Service
2	Sales
3	Deployment
4	Product
5	Technology Groups (later also Products and Projects)
6	Development
7	Research



This ordering is part of the model's presentation and should remain intact when the functions are represented or analyzed.

2.2 Functions are organizational building blocks

A function is not necessarily a separate legal entity, department or traditional cost center.

The model allows an organization to:

- have a particular function;
- not have a particular function;
- combine activities that might elsewhere be separate functions;
- have multiple organizational units performing activities associated with a function;
- allocate activities between functions according to the organization's actual operation.

The model therefore describes **what the organization does**, rather than prescribing how management must organize it.

2.3 Activities inside functions

The economic size of a function is determined by the activities belonging to it.

A function can therefore contain many activities with different:

- quantities;
- prices;
- costs;
- scopes;
- relationships to other functions.

The function-level result is a summary of the activities contained within that function.

2.4 Organization model with functions and product build scope areas

INCOME									
(payment from customer / market)									
v									
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+									
Organization	IPR	Functions	PRODUCT BUILD SCOPE AREA					response time of	
			(Products)					function	
			+-----+-----+-----+-----+-----+						
			P1	P2	P3	P4	P5		
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+									
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+									
		Support (HelpDesk)	1	1	1	1	1	online	
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+									
		Sales	1	1	1	1		online	
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+									
	DEC =	Deployment	1	1	1	0,5		online	
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+									
Production Management	Documenta-								
	tion,	Product areas					End-	1h	
	Education,						User		
	Competence								
							App-	4h	
							lica-		
						App-	tion	1d	
						lica-			
					Plat-	tion			1w
			Suppor	form	Plat-				
			t Func		form			1y	
			tions					1y	
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+									
		Technology Groups	1	1	1	1		6months	
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+									
		Development	1	1	1	0,5		1	
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+									
		Research	1	1				2y	
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+									
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+									

Figure 1 - This establishes **what the organization consists of** and preserves the function hierarchy.

3 Function as an Economic Summarizer

3.1 Fundamental function balance

Each function can be treated as a small GMVA summarizer.

The basic economic relationship is:

Income → Function → Purchase Costs + Refining → Profit/Loss

The direction of the funds is essential.

Income

Income/payment flows **into** the function.

Purchase Costs

Purchase Costs drain funds from the function.

They are not treated as funds entering the function's result.

Refining

Refining represents the expenditure associated with the activity through which the function performs its value-adding work.

Refining therefore also drains funds from the function.

Profit/Loss

The remaining economic result is:

$\text{Profit/Loss} = \text{Income} - \text{Purchase Costs} - \text{Refining}$

A negative result indicates that the function consumes more funds than it receives and therefore requires funding within the larger economic structure.

3.2 Internal summarization

The individual activities of a function can first be summarized and then the resulting function can participate in the larger organizational calculation.

This creates a recursive principle:

Activities → Functions → Organization

The same basic economic logic can therefore be used at different organizational levels.

3.3 Economic relationships between functions

Functions are economically connected to their neighboring functions.

A function may receive economic value or income from another function and may purchase something from another function. Consequently, the organization is not merely a collection of independent cost centers.

The functions form an **internally connected economic structure** whose individual balances contribute to the overall organizational balance.

3.4 Economic relationships between functions

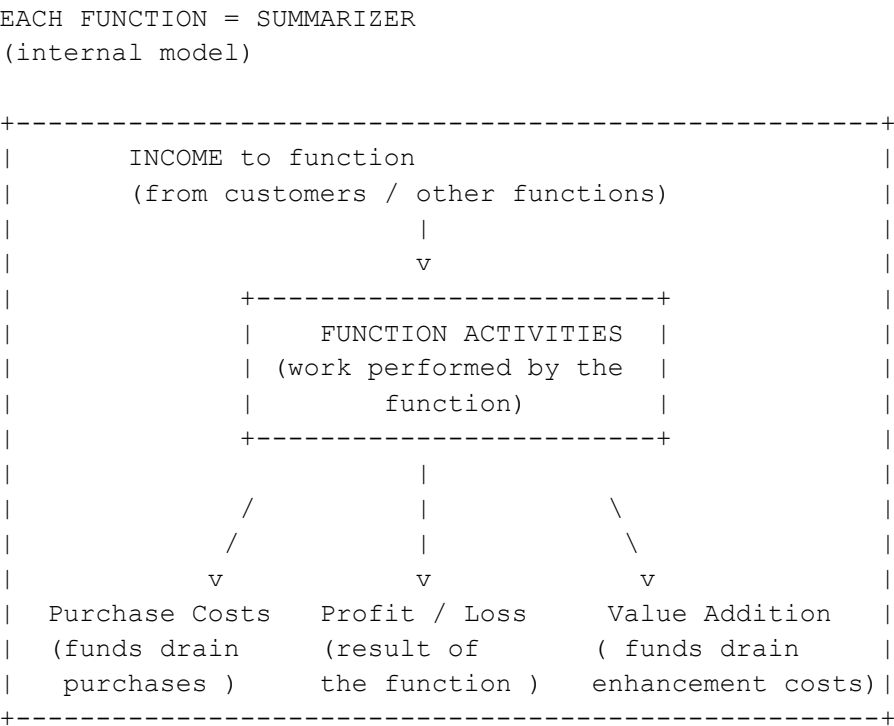


Figure 2 - Function as an Economic Summarizer

4 Product Build Scope Areas and Allocation

4.1 Product Build Scope Area

The original organizational model contains Product Build Scope Areas represented by the scope columns:

- Foundings
- Platform
- Application-Platform
- Application
- End-Customer

These are structural scope areas rather than definitions of what a particular organization must produce.

The important relationship is between a **function row** and a **scope column**.

A numeric value in a Product Build Scope Area means:

The function on that row participates in the scope identified by the column name.

The numeric value therefore describes the function's participation in that scope.

4.2 Scope areas can have quantities

The scope areas are not limited to a binary yes/no interpretation.

A scope area can contain an actual **number of instances**.

This makes the scope matrix useful as a calculation basis. For example, if there are:

- 2 Foundings;
- 5 Platforms;

then the model can recognize the resulting scope quantity when calculating the economic burden of a participating function.

4.3 Allocation through scope participation

A function's total economic result can be related to the scope areas in which it participates.

For example, where Research participates in Foundings and Platforms, and the relevant scope contains 2 Foundings and 5 Platforms, the resulting number of relevant Platform instances can provide an allocation basis.

If Research has:

- total cost: 100,000;
- income: 0;
- 10 relevant Platform instances;

then the allocated Research cost is:

$100,000 / 10 = 10,000$ per Platform

The total Research cost remains 100,000. Allocation does not create or remove cost; it determines **where the economic burden is attributed within the organizational/product-scope structure**.

4.4 Scope and economics

The Product Build Scope Area is therefore both:

1. a description of organizational/function participation; and
2. a potential calculation dimension for allocating economic values.

This is important for the intended practical implementation of the model.

5 Refining and Organizational Profitability

5.1 Refining

The term **Refining** is used for the value-adding activity represented by the model.

Refining does not require physical transformation.

The essential question is whether an activity produces an economically meaningful increase or improvement in value within the organization's structure.

The model therefore does not define value addition through the physical characteristics of an output.

5.2 Profitability adjustment

Because the function separately exposes Income, Purchase Costs and Refining, its profitability can be analyzed through different adjustment points.

The organization can investigate:

- how much income a function receives;
- how much it spends on purchases;
- how much it spends on Refining;
- how activities are allocated between functions;
- how activities participate in different Product Build Scope Areas;
- how changes affect function-level and organization-level Profit/Loss.

The objective is therefore not automatically to minimize every cost.

An additional expenditure may be economically justified when it creates sufficient additional value or useful reusable capability.

5.3 Reallocation of activities

An activity can potentially be analyzed under different functions.

This allows the model to examine organizational alternatives such as whether an expenditure should be associated more directly with a current operational activity or with Development or Research that creates capability used across several scopes.

This makes organizational structure itself an **economic analysis variable**.

5.4 Organization as the large summarizer

At the highest level, the complete organization applies the same principle as an individual function.

The organization receives external income, its functions consume funds through Purchase Costs and Refining, and the individual function results are summarized into the organization's Profit/Loss.

Thus:

Function = economic summarizer

and:

Organization = large economic summarizer composed of functions.

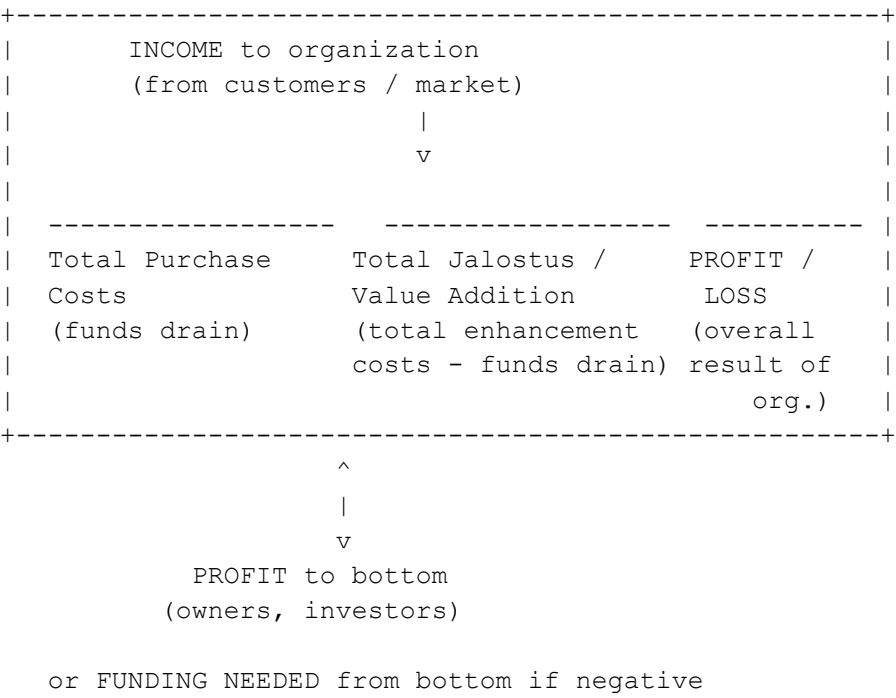


Figure 3 - the organization is itself a single summarizer built from its functions.

6 Historical Use Case: Hosting Business

The model was originally developed in connection with **IT hosting business**, and the first documented presentation in 2001 was titled *Hosting Business*.

Hosting is therefore the **historical environment in which the model was invented**, not a limitation or foundation of the model.

The hosting-business work demonstrated how the function structure could be used to analyze organizational profitability and the economic allocation of activities.

One important use was examining whether costs should remain directly associated with existing operational/product activities or instead be associated with **Development and Research**, where appropriate, when those activities created reusable capabilities supporting additional product variations.

The significance of this historical use case is therefore methodological:

The model could be used to investigate whether an alternative allocation of organizational activities and expenditure produced a better relationship between costs, value creation, product-scope support and profitability.

Historical Development and Hosting Business

1. 15 August 2001 — Original working model
 - Business-theme.xls
 - original organization/function structure
 - product-build structure
 - economic summarizer
 - response-time structure
2. 2001–2003 — Illustrative improvement scenarios
 - Budget 2001
 - Budget 2002
 - Budget 2003
 - explicitly labelled speculative/optimistic plans
 - not historical financial evidence
3. 2010 — Formal documentation
 - Yleinen lisäarvontuottamisen malli
 - formal research/documentation of the model

```
15 Aug 2001
ORIGINAL WORKING MODEL
|
+-- structure
+-- functions
+-- scope
+-- economics
|
+-- speculative improvement scenarios
|    2001 → 2002 → 2003
|
v
22 Jan 2010
FORMAL MODEL DOCUMENTATION AND CONCEPT OF SUMMARIZER
|
v
8 Sep 2026
COMBINED FUNCTIONALITY TO CREATE THE GENERIC MODEL
```

Figure 4 – Slow pace of progress to enhance model presentation

7 Application of the Summarizer Method

7.1 2001–2003 Improvement Scenarios

Document status: Source-derived historical analysis.

The 2001–2003 figures are **speculative planning scenarios**, not historical financial results or source-of-truth accounting data.

7.2 Purpose of the Historical Scenario

The original work dated **15 August 2001** provides an early practical example of the Generic Model of Value Addition being used not only to describe an organization, but to examine **how changes in the economics of its functions could improve the organization as a whole**.

The original material contains an organization/function structure together with economic calculations for 2001, 2002 and 2003. The organizational structure identifies functions such as Support, Sales, Deployment, Application, Application Platform, Platform, Support Functions, Technology Groups, Development and Research.

The three budget sheets should **not** be interpreted as historical financial accounts. They represent an optimistic, speculative development path: a scenario for how the hosting organization could potentially improve its economic result by changing quantities, unit economics and the allocation of activity between functions.

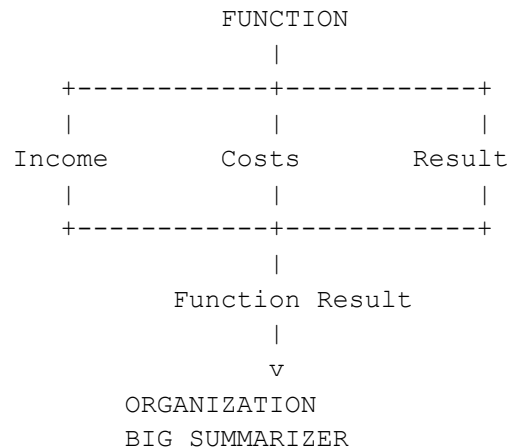
The important historical evidence is therefore not whether the forecast was subsequently achieved. The important evidence is that the model was being used as a **management and organizational improvement instrument**.

7.3 The Summarizer View

The central idea demonstrated by the original calculations is that each organizational activity can be viewed as a **summarizer**.

A function receives economic activity, performs its activities, consumes resources and produces an economic result. The individual results can then be summarized into the result of the whole organization.

Conceptually:



The original budget tables make this concept quantitative. They contain **Category, Quantity, Unit and Amount**, allowing the economic effect of changing the quantity or unit economics of an activity to be reflected in the organization-level result. The 2002 table, for example, contains Deployment, Application, Application Platform, Platform, Support Functions, Technology Groups and Development with quantities, unit values and resulting amounts.

This is an important extension of the purely structural organization diagram: the functions are not only organizational boxes. They become **economic summarizers whose changing values affect the summarizer above them**.

7.4 The 2001 Starting Scenario

The 2001 scenario represents the starting point of the hypothetical improvement exercise.

The surviving source material shows an organization-level result of approximately:

–336,000

The major positive contribution comes from Application activity, while several other functions and activity areas carry negative amounts. The source material shows, among other values, Application at 9,600 × 340 with a resulting **1,564,000**, Application Platform at **–375,000**, Platform at **–300,000**, Support Functions at **–500,000**, Technology Groups at **–250,000**, and Development at **–375,000**.

			income / â	units	costs / â	summary
Production Management	CED= Documentation, Education, Development	Support (HelpDesk)				
		Sales		500	500	-250 000
		Deployment	1 000	150	4 500	-525 000
		End-User		300 000		
		Application	9 600	340	5 000	1 564 000
		Application Platform		5	15 000	-75 000
		Platform		15	20 000	-300 000
		Support Functions		10	50 000	-500 000
		Technology Groups		50	5 000	-250 000
		Development				
		Research				
						<u>-336 000</u>

Figure 5 — Original 2001 economic summary chart / Excel chart

Key findings: The 2001 scenario represents the initial loss-making organization. Application is already a significant positive contributor, while several infrastructure, support and development-related activities consume more resources than their direct summarized contribution.

7.5 Changing the Summarizers: 2001 → 2002

The 2002 scenario changes the economics of several activities rather than simply removing costs indiscriminately.

For example, Deployment is represented as:

Quantity 1,000

Unit 250

Amount -375,000

while Application remains at 340 units but its unit cost changes from the 2001 scenario's 5,000 to 2,500, producing:

Application result = +2,414,000

Application Platform changes to a quantity of 25 with a resulting **-375,000**. Development is represented as three units at 125,000, producing **-375,000**.

The resulting organizational summary is approximately:

+164,000

as shown in the original source material.

		income / â	units	costs / â	summary
Production Management	Support (HelpDesk)				
	Sales		750	100	-75 000
	Deployment	1 000	250	2 500	-375 000
	End-User		300 000		
	Application	9 600	340	2 500	2 414 000
	Application Platform		25	15 000	-375 000
	Platform		15	20 000	-300 000
	Support Functions		10	50 000	-500 000
	Technology Groups		5	50 000	-250 000
	Development		3	125 000	-375 000
	Research				

164 000

Figure 6 — Original 2002 economic summary chart / Excel chart

Key findings: The hypothetical changes move the organization from an overall negative result to a positive result. The improvement is produced by changing the economics and volumes of individual activities, particularly the economically significant Application activity.

7.6 The 2003 Scenario: Increasing Value-Producing Capability

The 2003 scenario goes further.

Application increases to:

Quantity 500

Unit 9,600

Cost 1,500

Result +4,050,000

The source table explicitly records the resulting **4,050,000** contribution.

Deployment is represented by 350 units at 250, resulting in **-87,500**, while Sales becomes **-50,000**. The infrastructure-related negative contributions remain substantial.

The important addition is that the scenario also contains significant Development and Research expenditure. The original source material shows the 2003 scenario together with an organization-level result of approximately:

+1,412,500.

		income / â	units	costs / â	summary
Production Management	Support (HelpDesk)	50	10 000	5	450 000
	Sales		1 000	50	-50 000
	Deployment	0	350	250	-87 500
	End-User		300 000		
	Application	9 600	500	1 500	4 050 000
	Application Platform		30	5 000	-150 000
	Platform		25	15 000	-375 000
	Support Functions		15	20 000	-300 000
	Technology Groups		10	50 000	-500 000
	Development		5	125 000	-625 000
				1 000	
	Research		1	000	-1 000 000
					1 412 500

Figure 7 — Original 2003 economic summary chart / Excel chart

Key findings: The hypothetical organization becomes substantially more profitable even though significant resources are allocated to Development and Research. The scenario therefore demonstrates that a function can consume substantial resources while still being part of an economically positive organization.

7.7 Original Excel Charts and Cost Views

The original material also contains graphical representations of the economic scenarios, including cost-oriented charts and graphical summaries for the different years. The source reconstruction identifies separate cost views for 2001, 2002 and 2003.

These original graphics should be retained where available because they are evidence of **how the model was originally presented**, rather than merely being decorative illustrations.

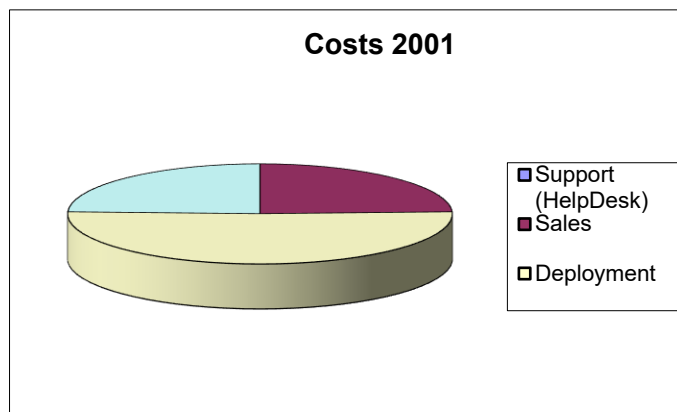


Figure 9 — Original 2001 cost distribution / pie chart

Key findings: Shows the relative distribution of modeled costs between the organizational activity categories in the 2001 scenario.

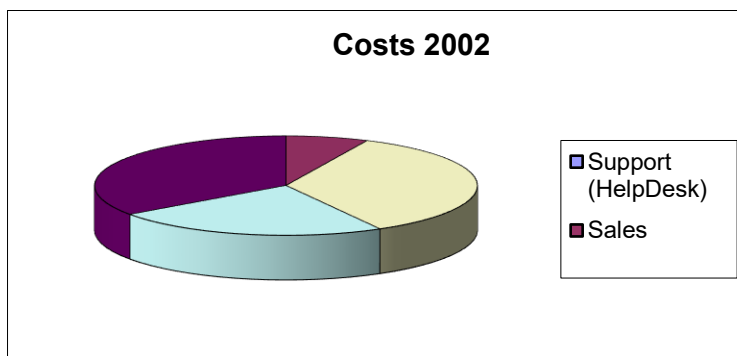


Figure 10 — Original 2002 cost distribution / pie chart

Key findings: Shows how the modeled cost structure changes in the 2002 improvement scenario.

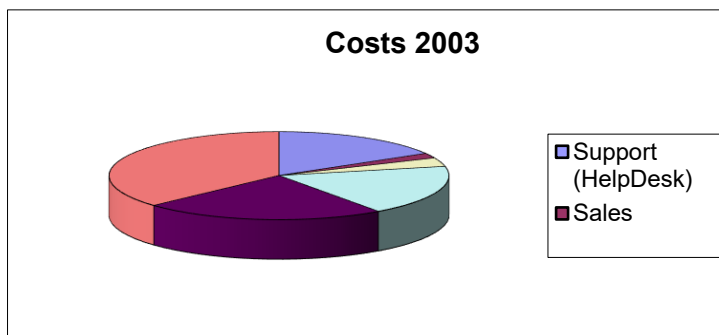


Figure 11 — Original 2003 cost distribution / pie chart

Key findings: Shows the modeled cost structure after the proposed 2003 changes, including the continued resource requirements of Development and other functions.

7.8 The Three-Year Trend

The three scenarios illustrate the intended management logic particularly clearly:

Scenario Organization result Interpretation

2001	-336,000	Hypothetical loss-making starting point
2002	+164,000	Hypothetical transition to profitability
2003	+1,412,500	Hypothetical substantially improved profitability

These values should be read as **scenario outputs**, not as historical performance measurements.

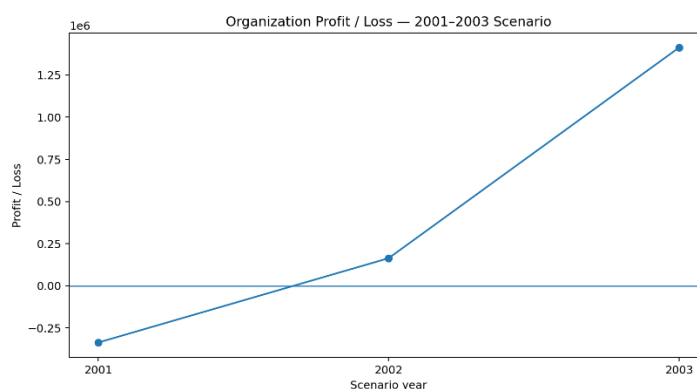


Figure 8 — 2001–2003 organization-result trend

Key findings: The modeled organization moves from a negative result in 2001 to a positive result in 2002 and a substantially larger positive result in 2003. The trend illustrates how changing the economics of individual summarizers changes the result of the organization-level Big Summarizer.

7.9 What the Scenario Says About Function Profitability

One of the most important findings is that individual function profitability is not necessarily the objective.

A function can have a negative direct economic result and nevertheless be necessary for the organization to create greater overall value.

This becomes especially visible in the 2003 scenario, where Development and Research consume significant resources while the organization simultaneously increases the contribution of Application.

The model therefore allows two different questions:

1. What is the economic result of this function?
2. What is the economic result of the organization when this function is part of the whole system?

These questions must not be confused.

A negative function result does not automatically mean that the function should be removed. It may represent:

- necessary operational expenditure;
- support for another function;
- capability creation;
- future-oriented Refining;
- or another contribution whose economic effect appears elsewhere in the organization.

The original scenarios do not establish a causal relationship between Research/Development expenditure and the later Application result. They demonstrate a **planned combination of expenditures and expected improved economics**, not a proven causal experiment.

7.10 The Improvement Strategy in Verbal Form

The historical scenario can therefore be described as follows:

The organization begins from a situation in which several functions consume more economic resources than they directly recover. The model is used to examine how the organization could improve its overall result by changing the economics of its activities.

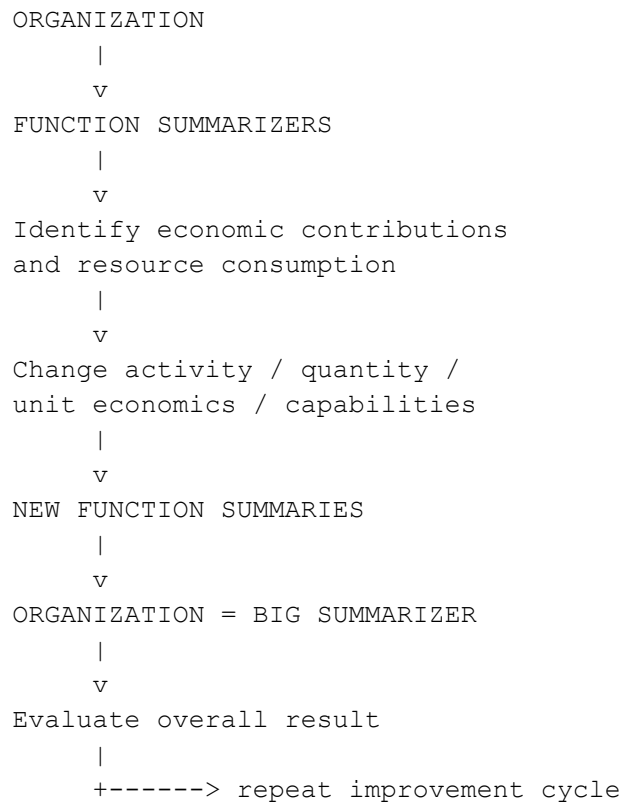
The proposed improvement is not simply a uniform reduction of costs. Instead, the economics of operational activities are improved, the cost of producing additional application activity is reduced, and the volume of the economically productive activity is increased. At the same time, resources can be allocated to Development and Research even though those functions have negative direct summaries.

The intended result is that the combined organization produces more economic value than it consumes. The individual function summaries are therefore intermediate measurements; the ultimate measure is the result of the organization as the Big Summarizer.

This is a more precise description of the historical material than treating the three sheets as ordinary annual budgets.

7.11 The Summarizer Method as an Iterative Management Process

The three scenarios also reveal a possible general management process implicit in the original work:



The model thus becomes more than a static profitability calculation.

It can be used as an organizational design and improvement model:

Describe → summarize → change → re-summarize → evaluate → iterate.

This interpretation is supported by the existence of multiple successive planning scenarios in the original 2001 work, but the actual achievement of the planned improvements is **not established by these documents**.

7.12 Significance for the Generic Model of Value Addition

The 2001 work provides an important historical foundation for the present model.

It demonstrates that the model can operate simultaneously at several levels:

FUNCTION

↓

FUNCTION SUMMARIZER

↓

FUNCTIONS COMBINED

↓

ORGANIZATION BIG SUMMARIZER

↓

ORGANIZATION RESULT

The 2001–2003 scenarios additionally demonstrate that the model can be used to examine **changes**, not merely a single organizational state.

This leads to a useful distinction:

Structural model

Describes:

- organization;
- functions;
- Product Build Scope Areas;
- participation;
- response-time characteristics.

Economic model

Summarizes:

- income;
- activity quantities;
- purchase costs;
- Refining;
- Profit/Loss.

Improvement model

Uses successive scenarios to examine:

- changing quantities;
- changing unit economics;
- changing activity costs;

- changing capability;
- changing resource allocation;
- resulting organization-level Profit/Loss.

The third layer is particularly relevant to the eventual application of the Generic Model of Value Addition.

7.13 Historical Source Status

For future work, the evidence should be classified explicitly:

Primary historical model evidence

- Original 15 August 2001 organization/function structure.
- Product Build Scope structure.
- Function and organizational summarizer concepts.
- Economic calculation structure.
- Original presentation charts.

Illustrative historical planning material

- Budget 2001.
- Budget 2002.
- Budget 2003.

The latter are **not source-of-truth financial records**. They are best preserved as examples of how the model was intended to be used to construct an optimistic improvement path for the hosting organization.

The 2010 documentation subsequently presents the model in a more formalized, industry-independent form. The reconstructed model explicitly defines the organization as a Big Summarizer and each function as a summarizer.

7.14 Key Finding

The most important historical finding is therefore:

The original Generic Model of Value Addition was not merely a method for calculating profitability. It was conceived as a way of representing an organization as a hierarchy of economic summarizers and then using changes in those summarizers to investigate how the economic result of the whole organization could be improved.

The 2001–2003 material provides an early concrete example of this principle:

individual economic changes → changed function summaries

→ changed organization summary → hypothetical organizational improvement.

This historical use case should remain separate from the generic definition of the model. The hosting business explains **where and how the model was originally applied**, but it does not define the scope of the Generic Model of Value Addition.

8 Product Build Scope Areas and Forked Production Structure

8.1 Product Build Scope Areas

A **Product Build Scope Area** defines a level or area of the production structure in which a Function participates in building or maintaining the organization's product.

The names

End-Customer → Application → Application-Platform → Platform → Foundings

are the original example used in the Hosting business model. They are **not intended as universal categories for all organizations**.

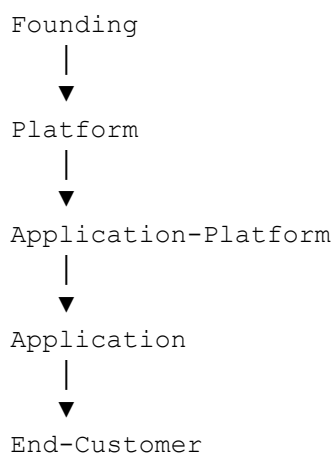
A different organization may require a different set of Product Build Scope Areas, with different names, different numbers of levels, or a different structure.

The important property is that the scope areas describe the organization's **business-specific production structure**.

8.2 From Linear Levels to a Forked Structure

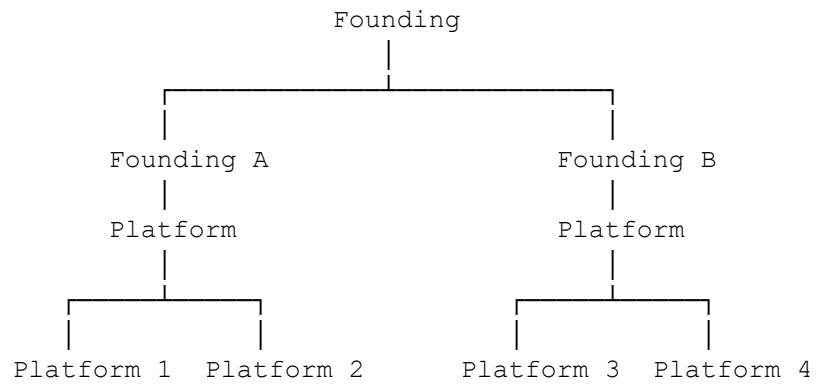
The scope areas should therefore not be understood as a single linear chain.

If an organization has only one instance at each level, the structure can appear linear:



When there are multiple instances of a scope area, however, the structure **forks**.

For example:



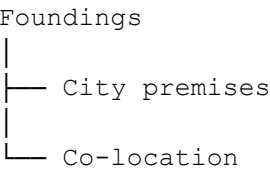
The same principle can continue at every subsequent level.

Thus, the Product Build Scope Areas form a **tree-like virtual production structure**, rather than a fixed five-level hierarchy.

8.3 Hosting Business Example

The original Hosting business provides a clear example of this structure.

There were two **Foundings**:



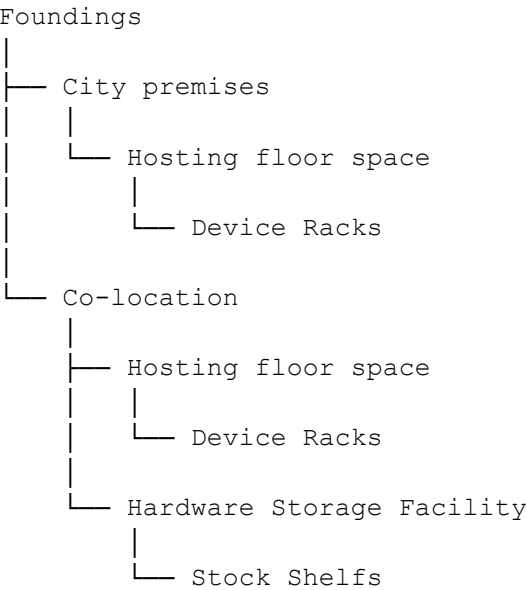
Both Foundings provided **Hosting floor space** as a Platform.

The Co-location Founding additionally provided a second Platform:



The next level then differs between those branches.

Both Hosting floor space Platforms provide **Device Racks** as Application-Platforms:

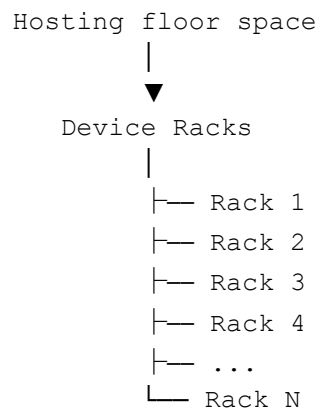


Thus, even though both branches use the same general Product Build Scope Area concept, their structures can diverge.

8.4 Applications Within the Branches - Hosting Floor Branch

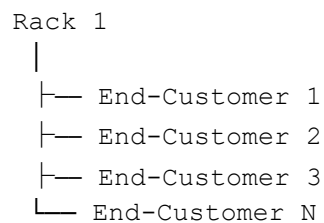
The next level can also fork into numerous individual Applications.

For Hosting floor space:



Each individual rack is an **Application** in this particular Hosting model.

Each rack can accommodate multiple End-Customers, for example server hardware:

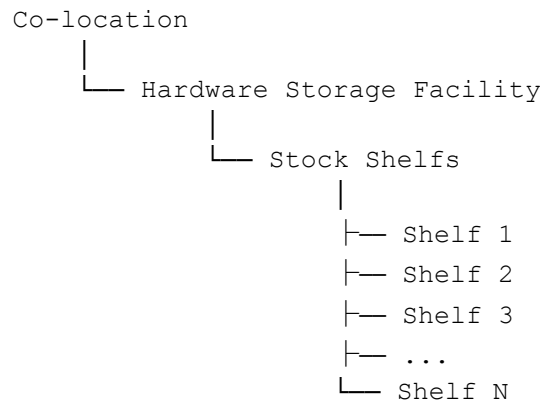


The same principle applies independently to every rack.

8.5 Applications Within the Branches - Storage Facility Branch

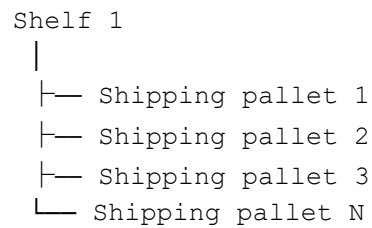
The Co-location Founding has another Platform: **Hardware Storage Facility**.

Its Application-Platform is **Stock Shelves**:



The individual shelves are the Applications in this branch.

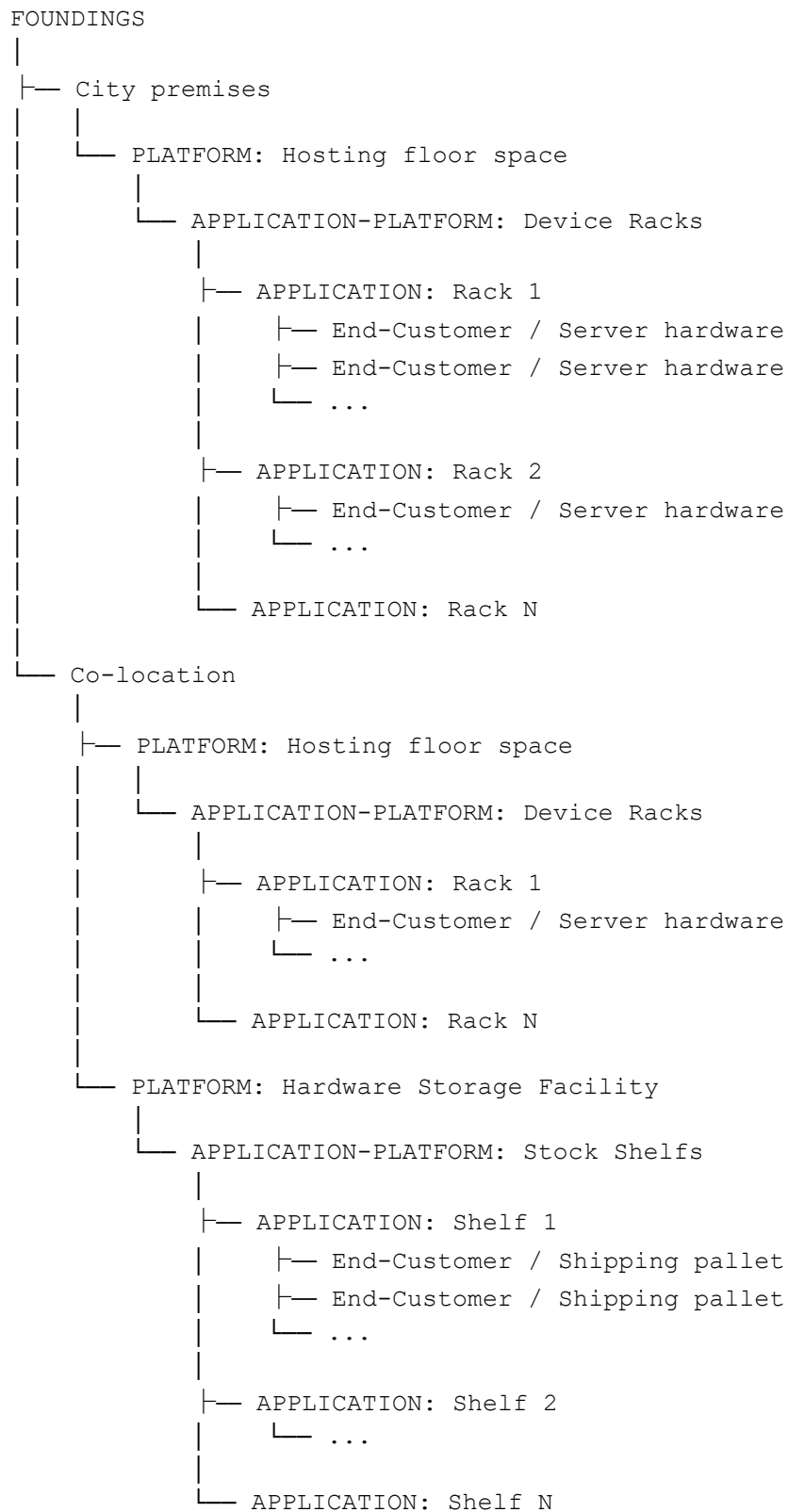
Their End-Customers are shipping pallets:



Consequently, the two branches have different physical and economic structures while remaining part of the same organization's Product Build Scope structure.

8.6 Complete Hosting Example

The complete structure can therefore be represented as:



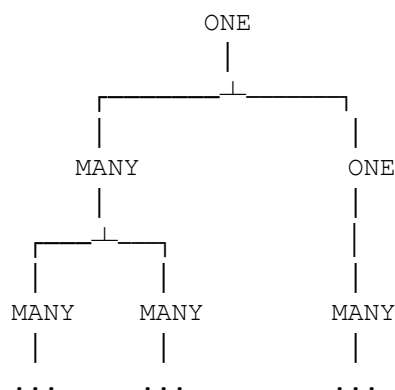
8.7 Forking Is a Structural Property

The significant property demonstrated by the example is **forking**.

A Product Build Scope Area can have:

- multiple instances;
- different numbers of instances in different branches;
- different lower-level structures in different branches;
- many Applications under one Application-Platform;
- many End-Customers associated with one Application.

Therefore:



The result is a **virtual tree of production capacity and relationships**.

The tree is virtual because it is a model of how the organization constructs and provides its product; it does not require that the organization's physical organization itself have the same tree structure.

8.8 Relationship to GMVA Functions

The Product Build Scope structure and the Function structure describe **different dimensions of the organization**.

The Product Build Scope structure describes **what is being built and through what production structure**.

The Functions describe **the economic activities performed by the organization**.

A Function can participate in one or several branches of the Product Build Scope tree.

For example, a Research Function may participate in the Foundings and Platforms of a Hosting business without being an Application or End-Customer itself.

This makes the Product Build Scope structure useful for allocating and analyzing Function economics across the production structure.

8.9 Generalization Beyond Hosting

The five names used in the original Hosting model should therefore be treated as an **example schema**, not as part of the universal GMVA definition.

The universal concept is:

An organization defines Product Build Scope Areas according to its own business. Multiple instances of a scope area can create branches, and those branches can independently contain further scope areas and instances, forming a tree-like virtual production structure.

This distinction is important for the later GMVA application: **the application should not hard-code End-Customer, Application, Application-Platform, Platform, and Foundings as universal levels.** They should be configurable according to the organization being analyzed.

9 Mathematical and Analytical Formulation

9.1 Basic mathematical GMVA function

The GMVA Function can be represented as an economic summarizer. Let:

- p = price charged per unit of the defined business transaction
- $q(p)$ = quantity of transactions demanded at that price
- c_P = Purchase Cost per transaction
- c_R = Refining cost per transaction

Income is:

$$I(p) = p \cdot q(p)$$

Purchase Costs are:

$$P(p) = c_P \cdot q(p)$$

and Refining is:

$$R(p) = c_R \cdot q(p)$$

The resulting Profit/Loss is therefore:

$$\boxed{\Pi(p) = I(p) - P(p) - R(p)}$$

or:

$$\boxed{\Pi(p) = q(p)[p - c_P - c_R]}$$

where Π represents the resulting **Profit/Loss**.

This provides a mathematical representation of the GMVA Function without requiring any particular industry, product or service.

The relationship can also be expressed as:

$$\boxed{I = P + R + \Pi}$$

where a negative Π means that the Function requires additional funding, while a positive Π means that the Function does not spend all of its Income.

9.2 The Function as an economic summarizer

The mathematical formulation expresses the internal economic principle of GMVA:

Income enters the Function. Purchase Costs and Refining go out of the Function. Profit/Loss is the resulting funding balance.

The Function therefore does not have to be interpreted as a physical production process.

Refining represents the economic expenditure associated with the activity through which value is added.

The Controller exists outside the Function. It may define or change parameters such as:

- the definition of one business transaction;
- Income per transaction;
- Purchase Costs;
- Refining;
- transaction volume;
- capacity;
- scope of the Function.

The mathematical model consequently provides the Controller with a way to evaluate the economic consequence of different parameter choices.

9.3 Economic feedback and control

Profit/Loss provides information about the funding state of the Function.

$$\Pi < 0 \Rightarrow \text{more funding needed}$$

$$\Pi > 0 \Rightarrow \text{Income is not fully spent by the Function}$$

The Controller can use this result when deciding whether the economic parameters of the Function should be changed.

Conceptually:

$$\text{Profit/Loss} \rightarrow \text{Controller} \rightarrow \text{changed economic parameters}$$

The Controller may change p , c_P , c_R , transaction volume, capacity, scope, or other parameters relevant to the particular Function.

This makes GMVA useful not only as a static profitability calculation but also as a framework for **iterative economic analysis**:

Describe → summarize → change → re-summarize → evaluate.

9.4 Verbal economic cases and conclusions

9.4.1 Case 1 — Price is too high

A Function may charge a high price per transaction:

$$p \uparrow$$

but the higher price may reduce demand:

$$q(p) \downarrow$$

Consequently, total Income may decrease despite the higher price:

$$I = pq(p) \downarrow$$

If Profit/Loss becomes negative, the conclusion is that the current economic parameters do not provide sufficient funding for the Function.

A possible Controller response is to reduce the price.

The resulting objective is not simply a lower price, but an improved overall Profit/Loss through a potentially higher transaction volume.

Conclusion: A higher price does not necessarily produce higher economic value. Price must be considered together with its effect on transaction volume.

9.4.2 Case 2 — Price is too low

A low price may increase demand:

$$q(p) \uparrow$$

and potentially increase Income. However, each transaction still generates Purchase Costs and Refining.

If these outflows consume too much of the Income, Profit/Loss may remain negative.

The Controller may therefore:

- increase the price;
- reduce Purchase Costs;
- reduce Refining expenditure;
- or combine these measures.

Conclusion: High transaction volume alone does not guarantee a positive economic result. Income must be sufficient in relation to Purchase Costs and Refining.

9.4.3 Case 3 — Purchase Costs are too high

The price and transaction volume may be appropriate, but Purchase Costs may consume too much Income:

$$P \uparrow \Rightarrow \Pi \downarrow$$

The Controller can investigate whether the required purchases can be obtained more economically.

Conclusion: A negative Profit/Loss does not identify the solution automatically. It identifies an economic problem that can be investigated through the individual components of the Function.

9.4.4 Case 4 — Refining expenditure is too high

The Function may have appropriate Income and Purchase Costs while spending excessive resources on Refining:

$$R \uparrow \Rightarrow \Pi \downarrow$$

The Controller may then investigate whether the same value-adding activity can be performed with less economic expenditure.

Conclusion: Refining is an economic variable that can be analyzed independently from Purchase Costs and Income.

9.4.5 Case 5 — Additional Refining can enable a higher price

Increasing Refining expenditure does not necessarily reduce economic value. Additional Refining may improve the value created by the Function sufficiently to make a higher price possible.

For example, additional Refining may improve the characteristics, capability, quality or scope of what the Function provides. If customers are willing to pay a higher price as a result:

$$R \uparrow$$

may enable:

$$p \uparrow$$

and potentially:

$$I = pq(p) \uparrow$$

If the resulting increase in Income exceeds the additional Refining expenditure, Profit/Loss improves.

Conclusion: Refining expenditure can be economically justified when it enables an increase in Income that more than compensates for the additional expenditure. Therefore, Refining should not be evaluated only as a cost to minimize; its potential contribution to increased value and Income must also be considered.

9.4.6 Case 6 — A function is individually loss-making

A Function may have:

$$\Pi < 0$$

without necessarily being economically unjustified for the organization as a whole.

Its expenditure may enable other Functions to generate additional Income or reduce their costs.

Therefore, a function-level negative result must not automatically be interpreted as a function that should be removed.

Conclusion: GMVA permits analysis at both the **Function level and Organization level**. A Function can be economically necessary even when its own summary is negative.

9.4.7 Case 7 — Organizational improvement through changing the structure

The same reasoning can be applied to the organization as a whole. The Controller can examine where resources are used, which Functions generate or consume economic resources, and how changes in one Function affect the economics of other Functions.

The objective is therefore not necessarily to minimize every individual cost. The objective is to improve the economic result of the **Organization as the Big Summarizer**.

Conclusion: GMVA provides a framework for evaluating organizational changes according to their effect on the combined economic result rather than according to isolated Function-level results.

9.5 General mathematical formulation

For a more general representation, let:

- x_t = current economic state of the Function;
- u_t = parameters controlled externally;
- $I(x_t, u_t)$ = Income;
- $P(x_t, u_t)$ = Purchase Costs;
- $R(x_t, u_t)$ = Refining.

Then:

$$\Pi_t = I(x_t, u_t) - P(x_t, u_t) - R(x_t, u_t)$$

The Controller can subsequently select new parameters based on the observed economic result:

$$u_{t+1} = C(\Pi_t, x_t)$$

and the Function may consequently enter a new economic state:

$$x_{t+1} = F(x_t, u_{t+1})$$

These equations describe an **external control relationship around the GMVA Function**, rather than adding Control as an internal input or output of GMVA.

9.6 Analytical significance

The mathematical formulation leads to several general conclusions.

1. **Profit/Loss is a funding balance.**

A Loss indicates that additional funding is required; a Profit indicates that the Function has not used all of its Income.

2. **Profitability depends on relationships, not individual variables.**

Price, transaction volume, Purchase Costs and Refining interact.

3. **A negative Function result does not automatically mean failure.**

The Function may support value creation elsewhere in the organization.

4. **The appropriate corrective action depends on the cause of the result.**

Price, Purchase Costs, Refining, volume, capacity or organizational scope may each require different responses.

5. **GMVA supports iterative organizational analysis.**

A proposed change can be evaluated by re-calculating the Function and then the Organization.

6. **The Organization is the larger economic summarizer.**

Function-level results provide intermediate economic information; the combined organizational result provides the broader measure of whether the organizational structure produces more economic value than it requires.

9.7 Relation to the computational analogy

The mathematical formulation also clarifies the previously discussed analogy with computational models.

The analogy should be understood as **structural rather than literal**. GMVA belongs to the economic domain, while the computational model belongs to the computational domain.

In GMVA:

$$\boxed{Income \rightarrow Function \rightarrow \begin{cases} \text{Purchase Costs} \\ \text{Refining} \end{cases} \rightarrow Profit/Loss}$$

The **Controller remains outside the Function**. Profit/Loss provides the economic information from which the Controller can determine whether additional funding is needed or whether Income remains after the Function's economic outflows.

This formulation preserves the central GMVA principle without introducing additional GMVA inputs or outputs.

9.8 GMVA as a Hierarchical System of Economic Summarizers

The GMVA Function does not necessarily represent only one business activity. A Function may consist of multiple individual economic units, and those units may themselves contain further levels of economic summarizers.

The resulting structure is **hierarchical**:

Summarizers can sit on top of other summarizers.

An upper-level Function can receive economic results from lower-level Functions, while also having its own Purchase Costs and Refining. The economic relationship between Functions therefore forms a nested organizational structure.

The 2003 planning case provides an illustrative example:

Function	Units
Support (HelpDesk)	10,000
Sales	1,000
Deployment	350
End-User	300,000
Application	500
Application Platform	30
Platform	25
Support Functions	15
Technology Groups	10
Development	5
Research	1

If each unit is interpreted as an individual economic summarizer, these rows represent:

$$10,000 + 1,000 + 350 + 300,000 + 500 + 30 + 25 + 15 + 10 + 5 + 1$$

311,936 individual economic units

Thus, for example, the **500 Applications** can be understood as approximately 500 separate individual summarizers at the Application level. In the 2003 context, an Application could represent an individual intranet or extranet application.

If Management and DEC are also represented by at least one summarizer each, the illustrated organization contains at least:

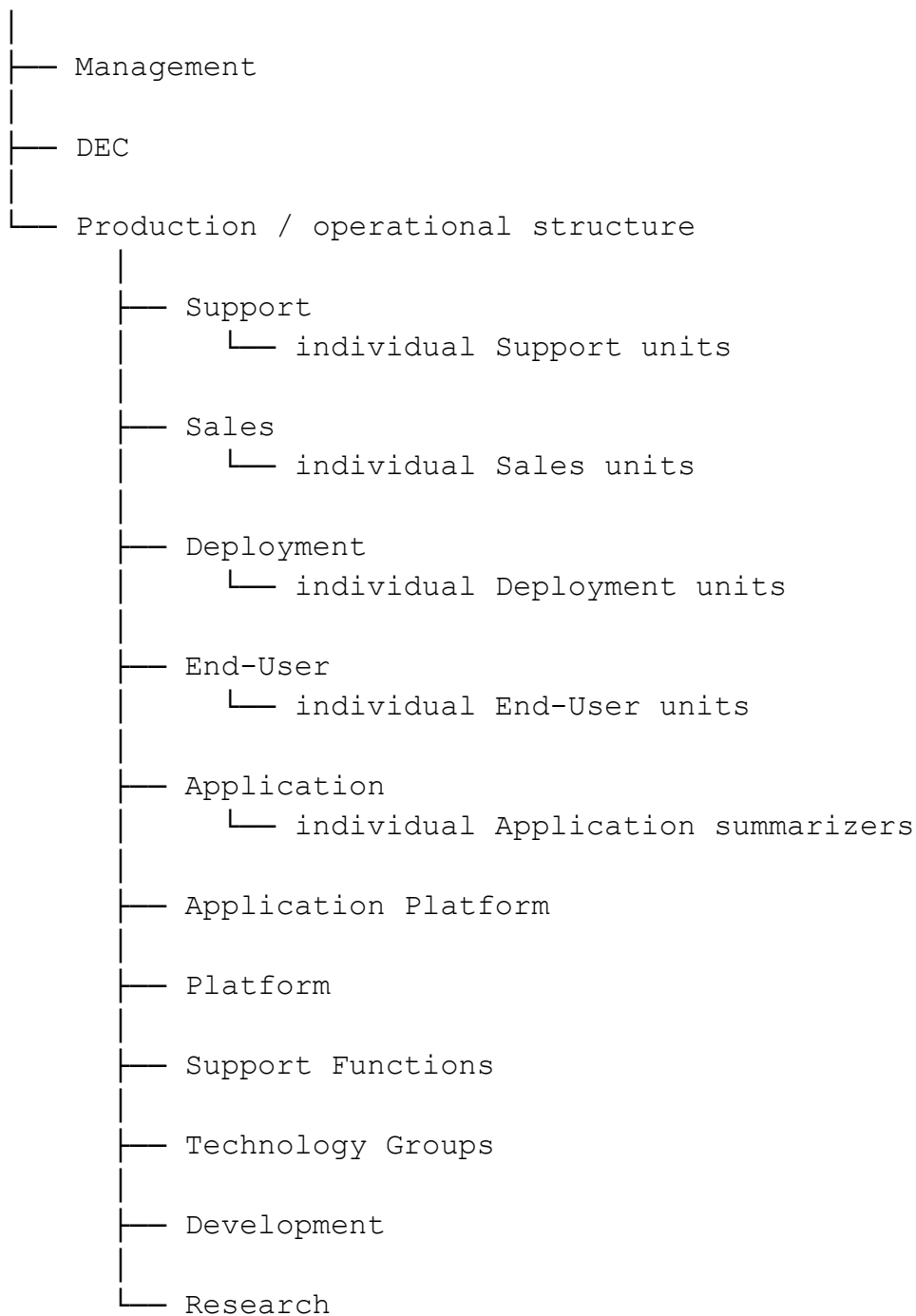
311,938 summarizer units

with Management and DEC participating across the organizational structure. The actual number may be higher because the 2003 table does not specify a unit count for Management or DEC.

9.9 Nested economic summarizers

The hierarchy can be represented conceptually as:

ORGANIZATION



The important property is that these are not necessarily independent, parallel activities. **A summarizer can sit economically above another summarizer.**

For example, a Sales activity may provide economic activity that enables Deployment. Development may receive funding from Research, while Development in turn enables Products and Applications. The exact relationships depend on the organization being analyzed.

9.10 Function-level mathematical aggregation

Let an individual economic unit j have:

$$\Pi_j = I_j - P_j - R_j$$

A Function containing n such economic units has:

$$\Pi_F = \sum_{j=1}^n \Pi_j$$

or:

$$\Pi_F = \sum_{j=1}^n (I_j - P_j - R_j)$$

Therefore:

$$\Pi_F = \sum I_j - \sum P_j - \sum R_j$$

This means that a Function such as **Application** can contain hundreds of individual Application summarizers while still producing one summarized Application-level economic result.

The same principle can then be applied upward:

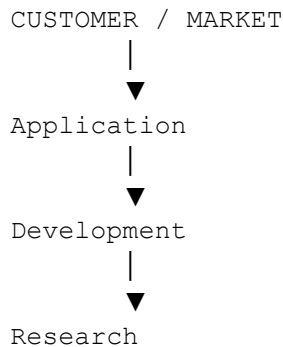
$$\Pi_{Org} = \sum_{F=1}^m \Pi_F$$

where the organization's Function-level summaries form the **Organization = Big Summarizer**.

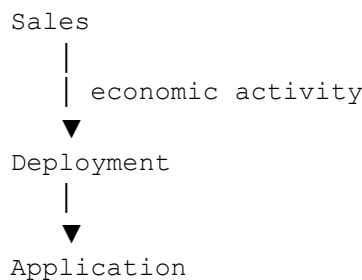
9.11 Internal economic relationships between summarizers

The hierarchy also explains why one Function can economically support another.

For example:



Similarly:



These relationships do not mean that every Function must itself produce a positive Profit/Loss.

A Function can have:

$$\Pi_F < 0$$

while its expenditure contributes to the ability of another Function to produce a positive result.

This is why the GMVA model can distinguish **local economic results** from the **overall organizational result**.

9.12 Management and DEC as cross-cutting summarizers

Management and DEC — Documentation, Education, Competence have a special structural position.

Unlike functions such as Sales, Deployment or Production, they do not necessarily have a corresponding Function above or below them in the same production chain.

Their economic activity can therefore support multiple other Functions simultaneously.

Their Purchase Costs can consequently be understood as **shared economic expenditure** allocated across the organization. In the conceptual model, the Purchase Cost arrows points outward to left from other functions towards the Management or DEC, so their costs may ultimately be covered by economic results generated elsewhere in the organization.

At the same time, Management and DEC remain ordinary GMVA Functions:

- they can receive Income from customers;
- they can receive **Funding from owners** where applicable;

- they have Purchase Costs;
- they have Refining expenditure;
- they produce a Profit/Loss result.

Their special characteristic is therefore **organizational position and shared participation**, not exemption from the GMVA economic model.

9.13 The organization as the complete summarizer

The complete organization can therefore be represented mathematically as a hierarchy:

$$\boxed{\text{Individual Summarizers} \rightarrow \text{Function Summarizers} \rightarrow \text{Organization Summarizer}}$$

At the lowest level:

$$\Pi_j = I_j - P_j - R_j$$

At a Function level:

$$\Pi_F = \sum_j \Pi_j$$

At the Organization level:

$$\boxed{\Pi_{Org} = \sum_F \Pi_F}$$

This expresses the fundamental GMVA principle:

The Organization is a Big Summarizer composed of economic summarizers, which may themselves contain further economic summarizers.

9.14 Important aggregation rule

The same economic transaction must not be counted twice.

If an Application-level summarizer is already included in the Application Function's summary, the Application Function summary is an **aggregation of those units**, not an additional independent economic result to be added once more to the same units.

Therefore, the organization can be viewed at different levels of resolution:

$$\boxed{\text{unit level} \rightarrow \text{Function level} \rightarrow \text{Organization level}}$$

while preserving the same underlying economic result.

This gives GMVA a potentially very large dimensionality: the 2003 example alone illustrates **311,936 quantified economic units**, before separately quantified Management and DEC units are added. The number is not a theoretical limit; it is determined by how finely the organization and its transactions are defined.

9.15 Conclusion from the 2003 case

The 2003 case therefore provides an important additional interpretation of GMVA.

GMVA is not merely a formula applied once to an organization. It can be applied **recursively and hierarchically**:

An economic activity can be a summarizer; a group of economic activities can be a summarizer; a Function can summarize its constituent summarizers; and the entire organization can summarize its Functions.

This makes it possible to analyze an organization at different levels of granularity while retaining the same fundamental relationship:

$$\boxed{\text{Income} - \text{Purchase Costs} - \text{Refining} = \text{Profit/Loss}}$$

The 2003 example consequently demonstrates how a relatively simple economic equation can describe an organization containing **hundreds of thousands of individual economic units**, connected through a hierarchy of Functions and summarizers.

The central analytical question then becomes not merely “**Is the organization profitable?**”, but also:

Where, at which level, and through which relationships is economic value being added, consumed, transferred, or left as a funding surplus?

This is a significant extension of the original single-Function formulation and fits directly with the idea that **GMVA is an organizational analysis method rather than merely a profitability calculation.**

10 Practical Applications

10.1 Financial-analysis application

One intended practical implementation is an application resembling an Excel-based GMVA calculation model.

Financial information could be taken from sources such as:

- accounting records;
- bookkeeping data;
- financial statements;
- annual reports;
- other organizational financial records.

The information could then be mapped into the GMVA organizational and economic structure.

The application could calculate and analyze:

- function-level Income;
- Purchase Costs;
- Refining;
- Profit/Loss;
- Product Build Scope allocations;
- organization-level profitability;
- relationships between costs and value-adding Functions;
- relationships between Functions and Product Build Scope Areas.

Product Build Scope Areas would be configurable for the organization being analyzed. The Product Build Scope structure has a **maximum of five steps**. The definitions of those steps are business-specific, while multiple instances at any step can create forks and therefore a tree-like virtual production structure.

This would turn financial data into an organizational value-addition analysis, rather than treating the financial statement only as a collection of accounting categories.

10.2 Organizational analysis

Jalostus by T. Lind can also be used to analyze:

- which Functions an organization actually performs;
- which activities belong to each Function;
- dependencies between Functions;
- internal economic relationships;
- distribution of costs;
- distribution of value-adding activity;
- organizational profitability;
- possible alternative organizational allocations;

- how the organization's Product Build Scope is structured and where it forks.

The Product Build Scope structure provides an additional view of the organization by showing how production is constructed from its business-specific scope areas.

10.3 Scenario analysis

Once Function quantities, activities, costs, Income and Product Build Scope participation are represented as data, the model can be used to compare alternatives.

Possible analyses include:

- changing the number of instances within Product Build Scope Areas;
- changing activity quantities;
- changing Purchase Costs;
- changing Refining expenditure;
- moving activities between Functions;
- changing the participation of Functions in Product Build Scope Areas;
- creating or removing branches in the Product Build Scope structure;
- analyzing different configurations of the five-step Product Build Scope structure;
- analyzing the resulting Function and organization Profit/Loss.

This allows the model to be used not only to describe an existing organization, but also to examine possible changes to its economic and production structure.

11 Further research

The original work concluded that the applicability and benefits of the model should be investigated more deeply through analysis of different types of organizations.

A natural continuation is therefore to test Jalostus by T. Lind against organizations with substantially different structures and purposes, rather than validating it only against its original Hosting-business context.

Particular attention should be given to defining precise calculation rules for:

- Function-to-Function economic flows;
- Product Build Scope allocation;
- multiple participation in Product Build Scope Areas;
- quantities and instances at each Product Build Scope step;
- branching of the Product Build Scope structure;
- activity-level aggregation;
- Function-level Profit/Loss;
- organization-level aggregation;
- treatment of negative balances and funding.

The Product Build Scope structure should be investigated as a **maximum five-step structure**, with the individual scope-area definitions determined by the business being modeled.

The long-term objective is a general-purpose computational implementation in which organizational financial data can be mapped to the model and used to analyze where resources are consumed, where Refining occurs, how value is distributed through the organizational structure and Product Build Scope structure, and how the resulting economics affect profitability.